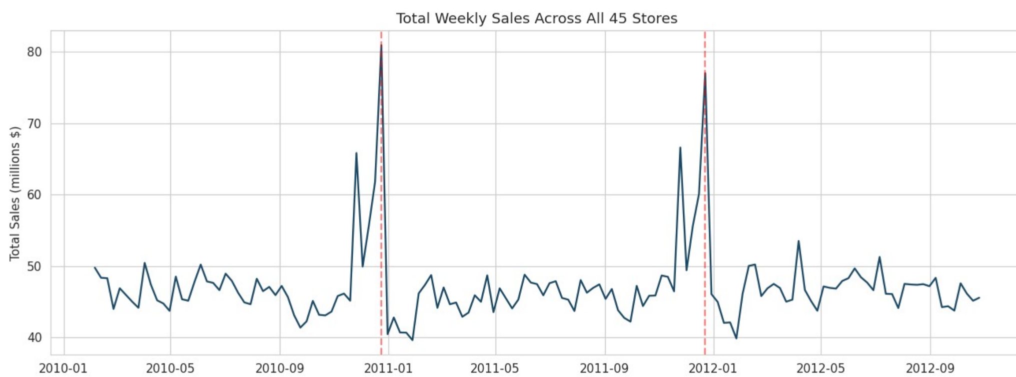
Walmart Sales Analysis

& 12-Week Forecasting with ARIMA / SARIMA

Data Science Capstone Project · Intellipaat · Problem Statement 1



Prepared by: Ankit Saxena

Dataset: 45 stores · 143 weeks · Feb 2010 – Oct 2012

June 2026 · Revised resubmission

Contents

1. Introduction & Problem Statement

A retail chain operating multiple outlets across the country is struggling to manage inventory — specifically, to match supply with demand. The objective of this project is to analyse three years of historical weekly sales data, convert it into actionable insight, then build models that forecast demand for each store over the coming weeks so inventory can be planned ahead.

The work is delivered in two parts, as set out in the capstone brief:

- **Part 1 — Analysis & Insights:** statistical analysis, EDA, outlier analysis and missing-value handling to answer six business questions on the effect of unemployment, seasonality, temperature and CPI, and to identify the best- and worst-performing stores.
- **Part 2 — Predictive Modelling:** segregate the data by store and forecast weekly sales for each of the 45 stores for the next 12 weeks using time-series models (ARIMA / SARIMA).

Revision note. This version revises Part 2 per reviewer feedback: the stores are explicitly segregated and forecast with seasonal time-series models (SARIMA), demonstrated in depth on five representative stores and then applied to all 45.

2. Dataset Overview

The dataset (walmart.csv) contains 6,435 rows and 8 columns — one record per store per week:

Feature	Description
Store	Store number (1–45)
Date	Week of sales (weekly, ending Friday)
Weekly_Sales	Sales for the given store that week (USD)
Holiday_Flag	1 if the week contains a major holiday, else 0
Temperature	Average temperature in the region (°F)
Fuel_Price	Cost of fuel in the region (USD/gallon)
CPI	Consumer Price Index
Unemployment	Regional unemployment rate (%)

Span: 45 stores across 143 consecutive weeks, 5 February 2010 to 26 October 2012. Weekly sales range from about $0.21M to $3.82M, averaging $1.05M.

3. Data Cleaning & Missing-Value Treatment

Data quality was checked before analysis. The dataset is complete and clean:

- **Missing values:** none — all 8 columns are fully populated across all 6,435 rows.
- **Duplicate rows:** none.
- **Type correction:** the Date field was parsed from dd-mm-yyyy text into a proper datetime and the records sorted by store and date to form a clean weekly time series for each store.

Because there are no missing values, no imputation was required. Had values been missing, the appropriate treatment would be time-aware interpolation within each store, since every store is an ordered weekly series.

4. Exploratory Data Analysis

Weekly sales are right-skewed: most weeks cluster around $1M, with a long tail of very large holiday weeks.

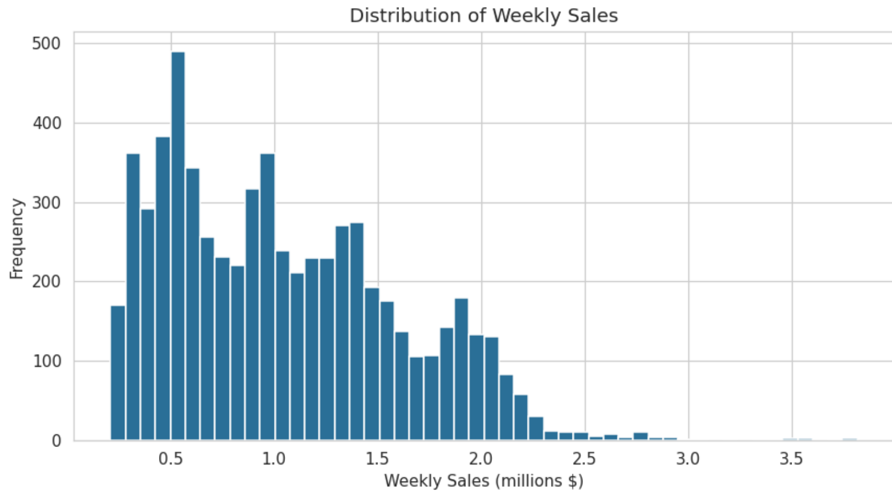


Figure 1 — Distribution of weekly sales across all stores.

A correlation matrix shows that, pooled across all stores, the external economic and seasonal factors are only weakly related to sales. The strongest associations are with unemployment and CPI, both mildly negative.

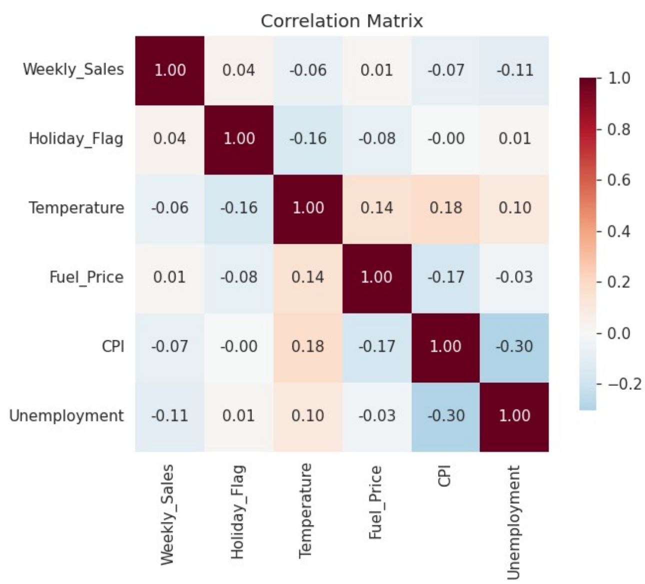


Figure 2 — Correlation between weekly sales and the economic / seasonal drivers.

Factor	Correlation	Interpretation
Unemployment	−0.11	Weak negative (strong for a few stores)
CPI	−0.07	Weak negative (strong for a few stores)
Temperature	−0.06	Negligible

Factor	Correlation	Interpretation
Holiday_Flag	+0.04	Small positive, but significant
Fuel_Price	+0.01	No meaningful relationship

The key message: macro variables are not strong linear predictors when all stores are pooled. Their real influence is store-specific and seasonal, which the next sections examine.

5. Outlier Analysis

Outliers in weekly sales were identified using the inter-quartile range (IQR) rule. Only 34 rows (0.53%) fall outside the fence, and the most extreme are all late-November and late-December weeks — the Thanksgiving and Christmas shopping peaks.

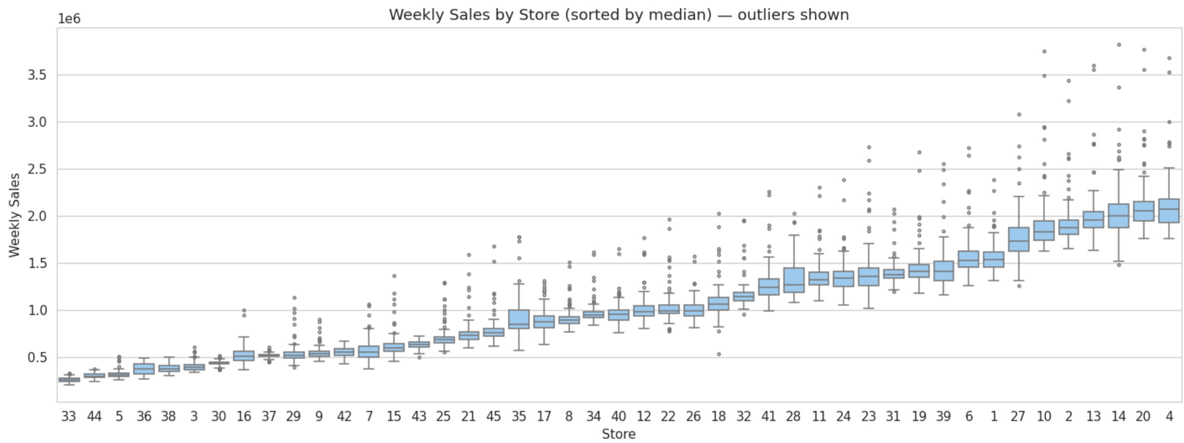


Figure 3 — Weekly sales by store (sorted by median); points beyond the whiskers are outliers.

Decision: outliers were retained. These are genuine seasonal peaks, not data-entry errors. Removing them would erase exactly the holiday-demand signal the business needs to plan for. They were therefore flagged and kept.

6. Business Insights

(a) Does unemployment affect sales — and which stores suffer most?

Across all stores the effect is weak and negative (r = −0.11), so unemployment is not a universal driver. But the relationship is highly store-specific: a handful of stores are very sensitive to local unemployment.

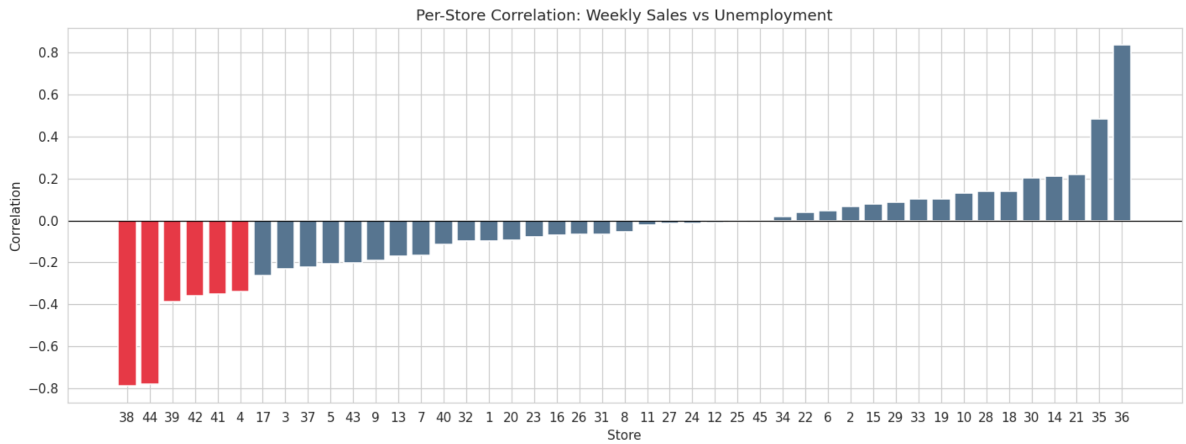


Figure 4 — Per-store correlation between weekly sales and unemployment.

Store	Corr. with Unemployment	Sensitivity
Store 38	−0.79	Very high
Store 44	−0.78	Very high
Store 39	−0.39	Moderate
Store 42	−0.36	Moderate
Store 41	−0.35	Moderate
Store 4	−0.34	Moderate

Answer: Stores 38 and 44 suffer most when unemployment rises, followed by 39, 42, 41 and 4. These should be protected with targeted promotions and leaner inventory during downturns.

(b) Is there a seasonal trend — when, and why?

Yes — a strong, repeatable annual cycle. Chain-wide sales sit around \$45–48M per week for most of the year, then spike sharply every November and December (Thanksgiving and Christmas).

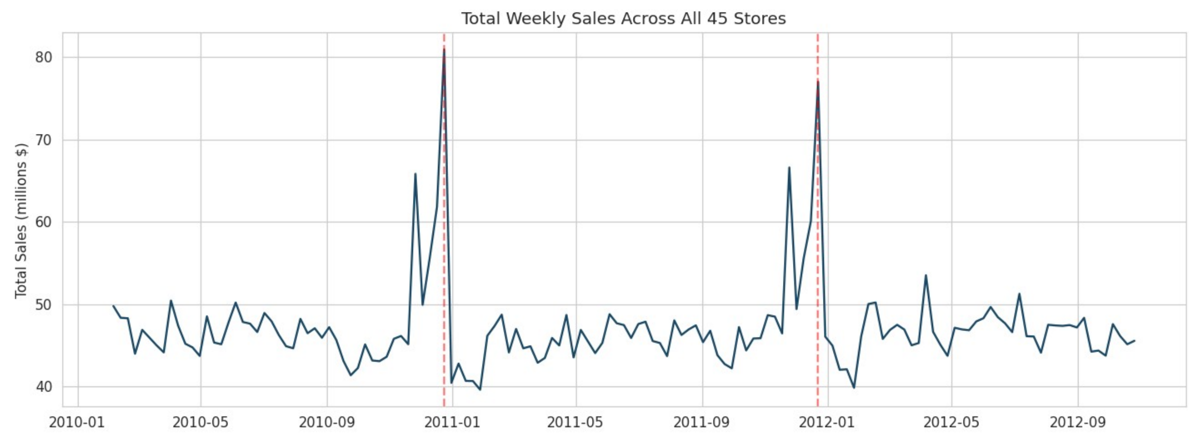


Figure 5 — Total weekly sales across all stores (red lines mark Christmas weeks).

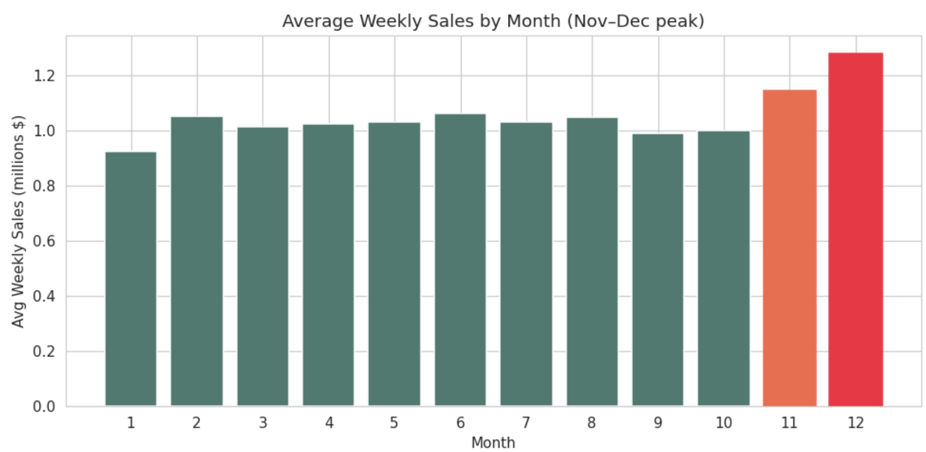


Figure 6 — Average weekly sales by month; November and December dominate.

(c) Does temperature affect sales?

Answer: Almost no effect (r = −0.06). Average sales are essentially flat across temperature bands, dipping only slightly in extreme heat. The calendar matters far more than the weather.

(d) How does the Consumer Price Index (CPI) affect sales?

Pooled, CPI is weakly negative (r = −0.07). As with unemployment, the impact is concentrated in particular stores.

Store	Corr. with CPI	Sensitivity
Store 36	−0.92	Very high
Store 35	−0.42	High
Store 14	−0.42	High
Store 30	−0.30	Moderate

Answer: CPI matters as a store-level factor rather than a chain-wide one. Stores 36, 35 and 14 are strongly price-sensitive, pointing to more price-conscious customer bases.

(e) Top-performing stores

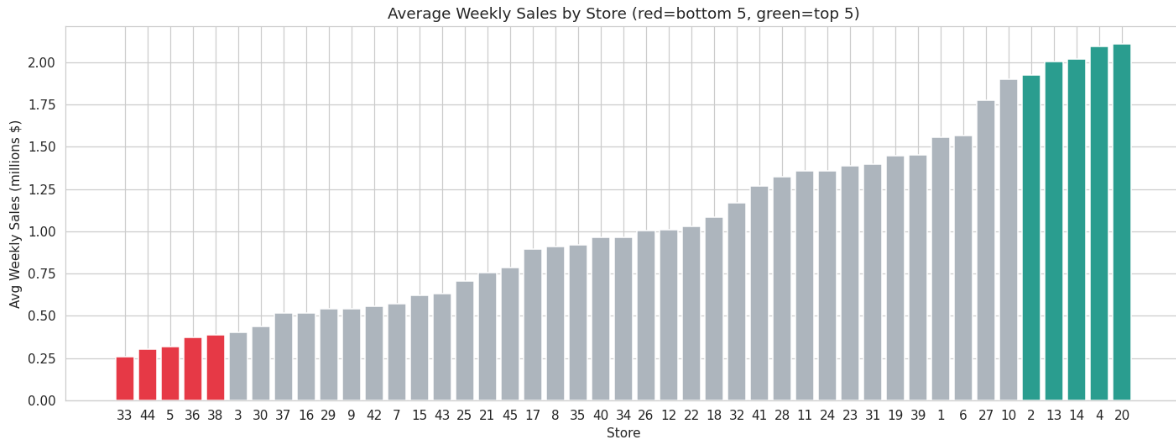


Figure 7 — Average weekly sales by store (green = top 5, red = bottom 5).

Store	Avg Weekly Sales	Total Sales (3 yrs)
Store 20	$2,107,677	$301,397,792
Store 4	$2,094,713	$299,543,953
Store 14	$2,020,978	$288,999,911
Store 13	$2,003,620	$286,517,704
Store 2	$1,925,751	$275,382,441

Answer: The top performers are 20, 4, 14, 13 and 2. Store 20 leads with ~$301M total. These flagships should be prioritised for stock availability, especially in Q4.

(f) Worst-performing store and the gap vs the best

Store	Avg Weekly Sales	Total Sales (3 yrs)
Store 33 (worst)	$259,862	$37,160,222
Store 44	$302,749	$43,293,088
Store 5	$318,012	$45,475,689

Answer: The worst performer is Store 33 (~$37.2M total). The best, Store 20, sells about 8.1× more — a gap of ~$264M (711% higher). A Welch t-test confirms the gap is highly significant ($t \approx 80$, $p \approx 3.5 \times 10^{-121}$), reflecting a real structural difference in store size/format/location. The smallest stores (33, 44, 5, 36, 38) are candidates for format review or right-sized inventory.

7. Predictive Modelling — Store Segregation & ARIMA / SARIMA

Inventory planning is a per-store problem, so the panel is segregated into one weekly-sales time series per store and each store is modelled individually. The 12-week horizon (Nov 2012 → Jan 2013) includes the holiday peak, so the model must capture the strong annual seasonality found in Part 1. We therefore use SARIMA (seasonal ARIMA, weekly period m = 52) and benchmark it against a non-seasonal ARIMA.

7.1 Segregating the stores

Each store becomes its own ordered weekly series (143 weeks, Friday frequency). The workflow is demonstrated in depth on five representative stores spanning the performance range — Store 20 (best), 19, 34, 9 and 33 (worst) — then applied to all 45.

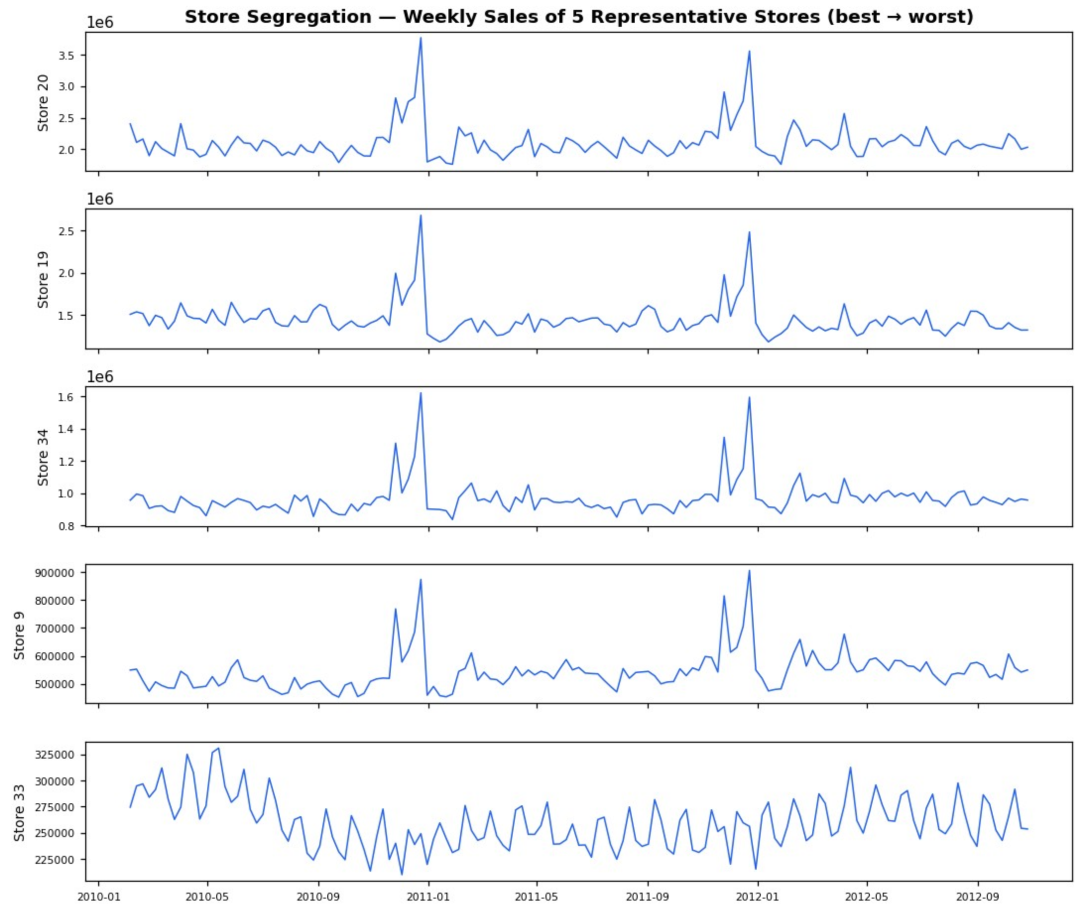


Figure 8 — Store segregation: weekly sales of five representative stores (best → worst).

7.2 Stationarity & model order

The Augmented Dickey-Fuller test rejects the unit-root null after first differencing (e.g. Store 20: ADF p ≈ 9×10^{-12} on the differenced series), so d = 1. The ACF of the differenced series shows a clear spike at lag 52 — the yearly seasonality — motivating the seasonal term. The chosen specification is SARIMA$(1,1,1)(0,1,1)_{52}$.

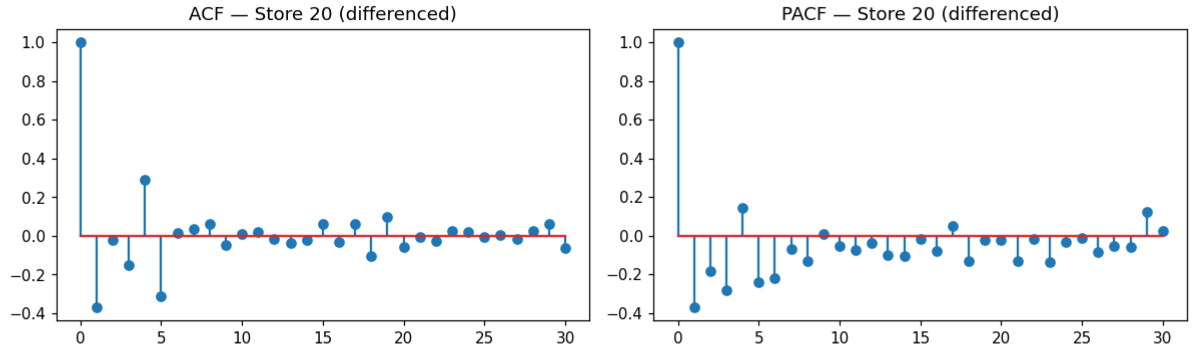


Figure 9 — ACF / PACF of the differenced series (Store 20).

7.3 ARIMA vs SARIMA

Adding the seasonal component is decisive. On Store 20 the AIC falls from 3,959 (ARIMA) to 963 (SARIMA) — a difference of nearly 3,000 — confirming that the yearly cycle is the dominant structure in the data.

7.4 Backtest — last 12 weeks held out

For each representative store the last 12 weeks were held out, forecast, and compared to actuals by MAPE. SARIMA is benchmarked against ARIMA on the identical hold-out.

Store	ARIMA MAPE	SARIMA MAPE
Store 20	4.09%	2.82%
Store 19	5.57%	2.75%
Store 34	3.08%	1.76%
Store 9	3.76%	3.29%
Store 33	2.50%	3.02%
Mean	3.80%	2.73%

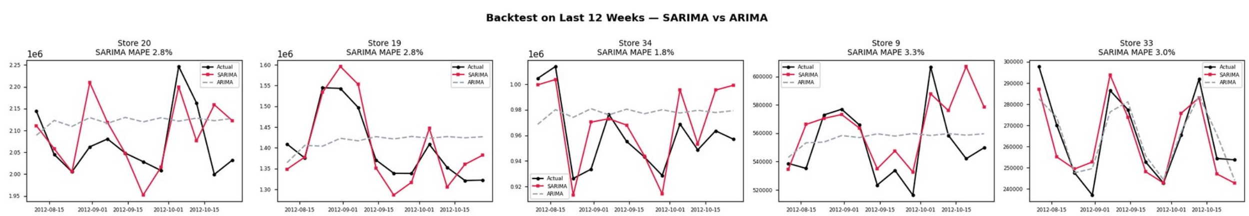


Figure 10 — Backtest on the last 12 weeks: actual vs SARIMA vs ARIMA.

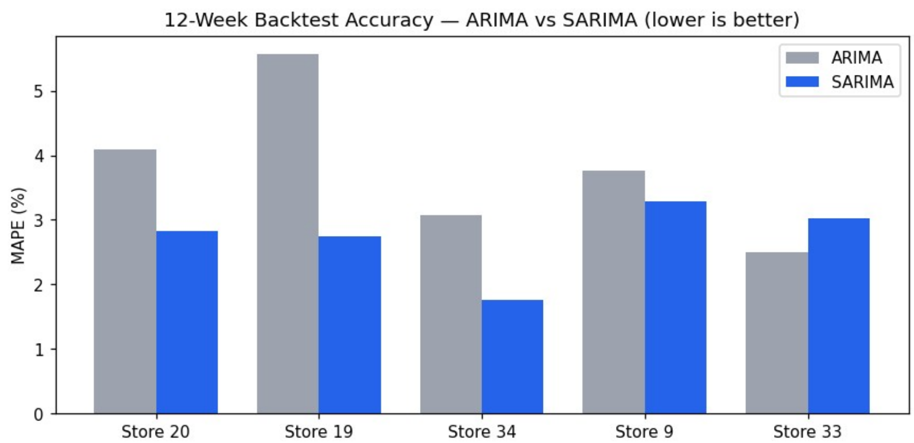


Figure 11 — 12-week backtest accuracy, ARIMA vs SARIMA (lower is better).

SARIMA wins. Its mean 12-week MAPE of 2.73% beats ARIMA (3.80%) and improves on the earlier Holt-Winters baseline (4.33%), while correctly tracking the holiday surge that flat ARIMA misses.

7.5 Forward 12-week forecasts

Re-fitting SARIMA on each store's full history yields the 12-week forecast with 95% confidence bands. The forecasts reproduce the Thanksgiving/Christmas peak, and the bands widen through the volatile year-end period.

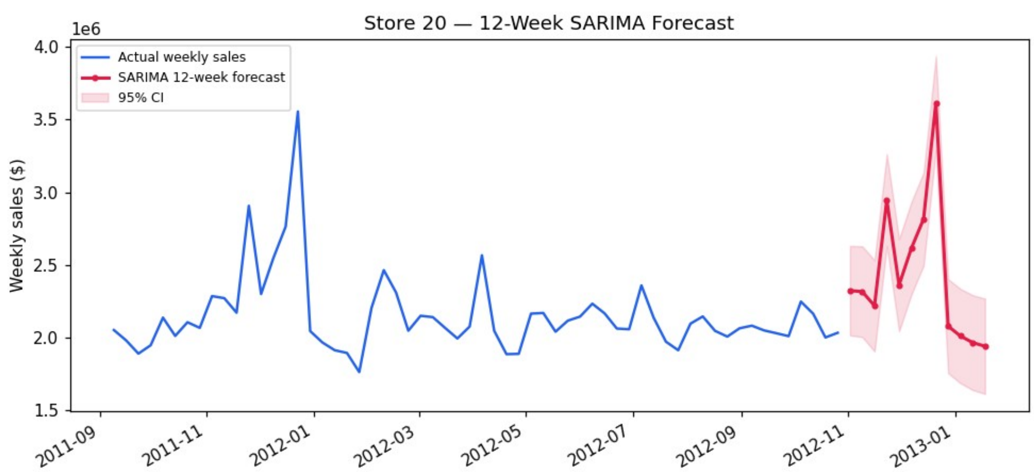


Figure 12 — Store 20 (best performer) — 12-week SARIMA forecast.

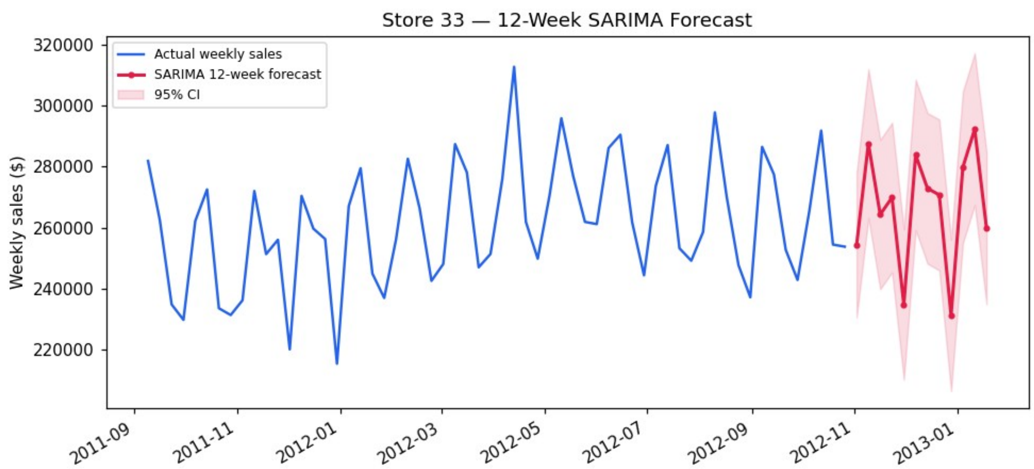
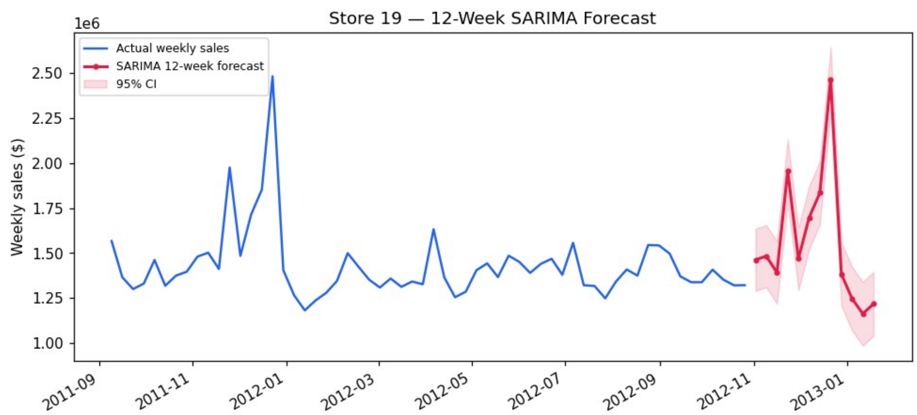
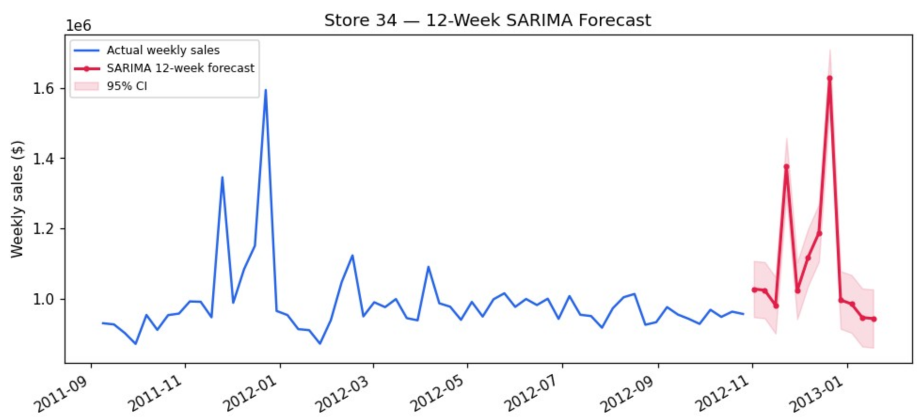
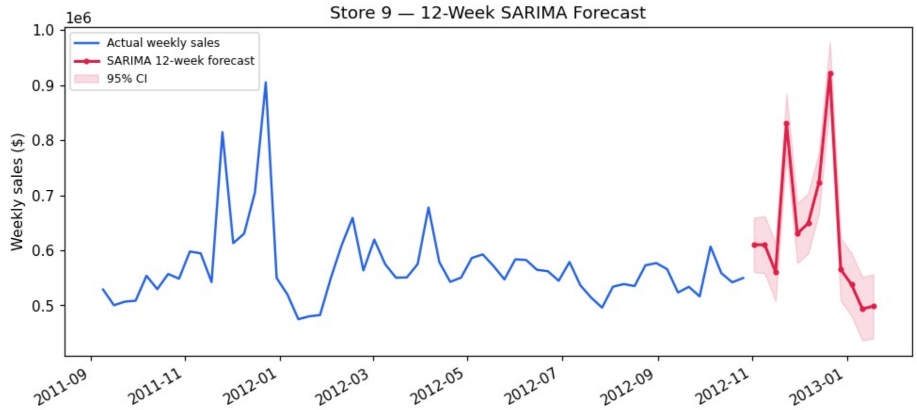


Figure 13 — Store 33 (worst performer) — 12-week SARIMA forecast.







Figures 14–16 — 12-week SARIMA forecasts for Stores 19, 34 and 9.

7.6 Forecasting all 45 stores

The validated SARIMA pipeline is fit per store across the whole chain, producing the required 12-week forecast for every store (45 × 12 = 540 forecasts, each with a 95% band), exported to walmart_12week_forecast.csv. The chain-wide 12-week outlook totals approximately $623.8M.

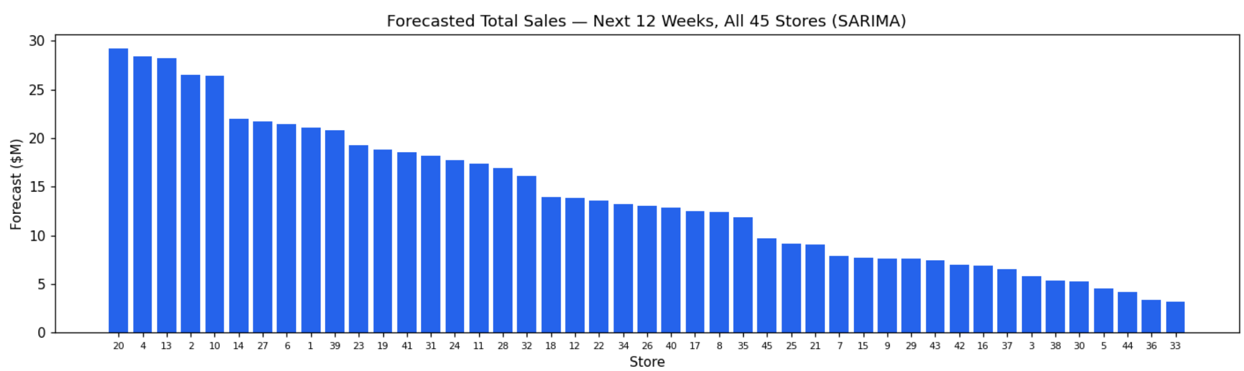


Figure 17 — Forecasted total sales over the next 12 weeks, all 45 stores.

Example — Store 20's forecast captures the holiday ramp:

Week ending	Forecast sales	Note
2012-11-02	$2,322,001	
2012-11-16	$2,266,757	
2012-11-23	$2,931,015	Thanksgiving
2012-12-21	$3,612,907	Christmas peak
2012-12-28	$2,089,182	Post-holiday

8. Conclusions & Recommendations

The analysis converts three years of weekly sales into a clear, actionable picture for inventory planning:

- Sales are driven by the calendar, not the weather — a strong Nov–Dec seasonal peak dominates, while temperature and fuel price are negligible.
- Macro factors (unemployment, CPI) matter at the store level, not chain-wide: stores 38 and 44 are unemployment-sensitive; 36, 35 and 14 are price-sensitive.
- Store performance is highly unequal — Store 20 sells ~8× Store 33 — a statistically significant structural gap that should guide store-level inventory and format decisions.
- **Forecasting (revised):** each store is forecast with SARIMA (seasonal ARIMA, m = 52), validated by a 12-week backtest (mean MAPE ≈ 2.7%, beating ARIMA and Holt-Winters). A 12-week forecast with confidence bands is produced for all 45 stores for inventory planning.

Recommendation: drive the replenishment plan from the per-store SARIMA forecasts — build holiday stock early for the flagship stores (20, 4, 14, 13, 2), right-size inventory for the smallest stores, and protect unemployment- and price-sensitive stores with targeted promotions.